IBM SkillsBuild AI
Experiential Learning Lab



Solution Design

Team Name: Void

Team Leader: Jettapol Tuetrakul

IBM

How to Use This Guide

Purpose: This guide offers a template for teams to capture essential information. It aims to help you move from week to week, completing essential exercises to deliver an innovative, user-centric solution at the end of week 10.

Important Notes:

‼️ Each slide has an indication of by when we recommend that you complete these assignments – however, you are free to advance or delay work according to your schedule.

‼️ Teams are expected to organized themselves, share tasks, find time to meet and complete the assignments and exercises.

‼️ Your hard deadline is the end of week 10 – by then, your team must submit a functioning MVP, project documentation and solution demonstration.

‼️ Submit your weekly deliverables in the forms: https://wkf.ms/4sDluV8

Tips:

✅ Keep language user-focused, avoid technical jargon, and write in short, clear sentences.
✅ Use images and illustrations – one image says more than 1000 words. Copy here images of your boards, designs, etc. or use illustrative images.
✅ Customize according to your needs – feel free to change or adapt the templates.

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Opportunity Framing

From industry problem to user-centered opportunity

Before completing the Opportunity Framing section, teams are strongly encouraged to first conduct two foundational exercises:

- Empathy Map – This helps you deeply understand the user's perspective, emotions, and motivations. It ensures that the opportunity framing is grounded in real user needs rather than assumptions.
- As-Is Scenario Map – This captures the current user journey, including steps, pain points, and context. It provides clarity on where friction occurs and why existing solutions fail.

Completing these exercises first will make the opportunity framing process more accurate and user-centered. They provide essential insights that inform sections such as Core User Tension, Current Experience, and Why Current Solutions Fail.

- Optional: Stakeholder Map – If you are finding it difficult to identify users and stakeholders, this exercise helps capture internal and external actors that might influence your solution.
- Optional: Research Plan - A research plan is essential when starting a project. It fits in especially well right after you identify your team's Assumptions and Questions.

Industry Problem Context

Purpose: Define the industry-level challenge and why it matters
When: By week 1

Tip:
- Refer to the Industry Problem Statements document.
- As a team, discuss and select a problem statement to tackle.
- In this slide, summarise what you know/understand about the problem. As you learn more and get deeper insights into the industry, update this slide.

Industry challenge statement

Government buyers must prove that every purchase is fairly priced and free of fraud. But price research and vendor checks are done by hand, which is slow and inconsistent, and citizens can rarely see how their money is spent.

Why this problem matters

- ***Impact****: The U.S. federal government alone spends more than $700 billion a year on contracts. Even small overpayments add up to large losses of public money.*
- ***Consequences****: Overpricing, bid rigging and shell companies go undetected. Purchases are delayed by manual research. Public trust drops when spending isn't transparent.*
- ***Strategic relevance****: Governments are under pressure to modernize, cut waste and be more open about spending.*
- ***Stakeholders****: Procurement officers, approving supervisors, auditors and inspectors general, honest vendors, and taxpayers.*

Who Is Experiencing the Problem?

Purpose: Identify the people most affected.
When: By week 1

Key tools:
- Empathy Map
- Stakeholder Map

Tip: To gather this information, you will need to carry out industry and user research.

- **Define Clear Objectives:** Start by identifying what you need to learn about = users—e.g., pain points, workflows, or feature expectations.
- **Choose the Right Methods:** Define the best way to gather the required information – desk research, user interview, survey, etc.
- **Document and Synthesize Findings:** Organize the information using key tools and list the key insights about the problem, user needs and pain-points, existing solutions, and opportunities for AI solutions.

Primary user

A procurement officer (contract specialist) at a city, state or federal agency who handles dozens of purchases a month. They must document that each price is reasonable and each vendor is eligible, often under deadline pressure.

User environment

- ***Setting**: an office desk with email, spreadsheets, PDF quotes and several government websites open at once*
- ***Tools and processes:** SAM.gov for vendor checks, USAspending.gov or internal records for past prices, web searches for market prices, Word or Excel for memos, an agency procurement or finance system for the final award*

Secondary stakeholders

- ***Approving supervisors:** sign off on large purchases*
- ***Auditors and inspectors general:** review decisions after the fact*
- ***Vendors:** honest vendors lose to fraudulent ones*
- ***Citizens and journalists:** want to know how public money is spent*

Core User Tension

Purpose: Capture the heart of the opportunity

When: By week 1

Key tools:
- Empathy Map
- As-Is-Scenario

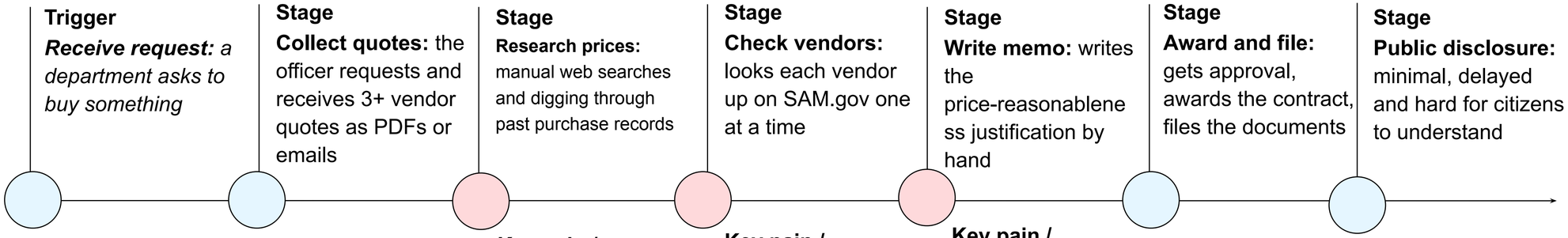
Tip: Focus on understanding what users are experiencing.
- Interview or survey users
- Consult with industry experts
- Explore industry reports and case studies

User goal

1 sentence: *What is the user trying to achieve? Keep it short and clear.*

Current User Experience As-Is Snapshot

Map the existing journey and key steps users take now



Key pains / frustrations

Max 3: *List the top fiction/pain points that block the user's goal.*

1. **Time:** price research and memos take hours per purchase.

2. **Hidden fraud**: collusion between vendors (shared addresses, bank accounts, staged bids) is almost impossible to spot by hand.

3. **Audit and transparency pressure:** documentation varies from officer to officer, and citizens get little visibility.

Key pain / frustration
slow

Key pain / frustration
easy to miss links between vendors

Key pain / frustration
rework, inconsistent quality

Where friction (pain points) occurs

Highlight where pain-points, such as delays, errors, rework, or cognitive overload happen.

Source: If applicable, describe source origin

Why Current Solutions Fail

Purpose: Understand gaps in existing tools/processes
When: By Week 1/2

Key insight – why the problem exists

The data needed to judge a purchase already exists, but it's spread across separate systems, and no tool connects it for a single purchase decision. The officer has to be the integration layer.

Key tools:
- Empathy Map
- As-Is-Scenario

Tip: Focus on understanding what solutions users are already using and why they don't solve the problem – this will help you understand what opportunities exist for you to innovate and offer better solutions.
- Research existing solutions
- Interview or survey users
- Consult with industry experts

Existing tools / processes used today

- *SAM.gov (vendor registration and exclusions list)*
- *USAspending.gov and FPDS (past contract data)*
- *Web searches and vendor catalogs*
- *Spreadsheets and Word templates*
- *Agency procurement and ERP systems*

What they do well

- Authoritative, official data sources
- Searchable records of past awards
- Familiar to officers and accepted by auditors

Where they break down

- *Each source is checked separately and manually*
- *No comparison of quote line items against past prices*
- *No detection of links between competing vendors*
- *Memos are written from scratch each time*
- *Public data is raw and technical, not written for citizens*

Early AI Opportunity Hypothesis

Purpose: Understand opportunities to offer better solutions (No solution design yet)
When: By Week 1/2

Why Agentic AI?

- ***Read*** *messy quote PDFs in any format*
- ***Match*** *products described differently ("Latitude 5440, i7, 16GB") to past purchases and market prices*
- ***Connect*** *evidence across sources (new vendor + shared address + 3× markup = high risk)*
- ***Adapt*** *the checks to each purchase and follow up when something looks wrong*
- ***Write*** *an audit-ready memo for officers and a plain-language summary for citizens*

High-level hypothesis

An intelligent system could help procurement officers by automatically reviewing vendor quotes against past government prices, market prices and vendor records, flagging risks with clear evidence and drafting audit-ready reports, because today this work is done by hand across separate systems, which is slow, inconsistent and misses fraud patterns between vendors.

Key tools:
- Empathy Map
- As-Is-Scenario
- Big Idea Vignettes

Tip: Focus on understanding how AI can offer value beyond basic prompting and automation – this will help you understand what opportunities exist for you to innovate and offer better solutions.
- Understand Agentic AI capabilities
- Consult with industry experts
- Research innovative Agentic AI solutions in other areas/industries that might give you inspiration

✅ Checkpoint: Week 2

At this stage, you should have a clear picture of the problem, the people affected, and why current solutions fail —without jumping into solution design.

- ❑ The problem is specific and human-centered (experienced by real people)
- ❑ The user and user experience is clear
- ❑ The pain-points are solvable and non-obvious
- ❑ There is a clear and valuable opportunity for Agentic AI to offer a better solution and experience

Key tools:
- Empathy Map
- As-Is-Scenario

Tip: There are things we know –there are things that we think we know (assumptions) – and there are things we do not know. Now it is the time to look again at your empathy map and as-is-scenario and do a critical analysis. Is there something that is critical for the success if your project that you might not know for sure yet?

User value is at the heart of innovation. The market is full of solutions that solve no-one's problem because they failed to understand their user needs, pain-points and desires – this is the time for you to be sure that you are solving a real, human problem, that will make users want to use your solution.

Open Questions & Assumptions

What we don't know yet

List gaps in knowledge that need validation.

1. *How long price research and memo writing actually take officers per purchase*
2. *Which fraud patterns officers worry about most*
3. *How much detail auditors expect in a price-reasonableness memo*

Assumptions we are making

Document any assumptions about users, processes, or technology.

1. *Officers would trust AI flags if each one shows its evidence*
2. *Past award data (USAspending) is detailed enough to compare prices*
3. *Agencies want to publish plain-language spending summaries*

Validation research

List what research is needed to gather information and/or validate assumptions

- ❑ *1. Review the FAR price analysis guidance (Part 15.404-1) and public GAO and inspector general audit reports*
- ❑ *2. Study DOJ Procurement Collusion Strike Force case examples to choose realistic fraud patterns*
- ❑ *3. If possible, interview 1–2 people with procurement or audit experience*
- ❑ *…*

User-Centric Solution

From opportunity to concept – addressing user pain-points

Before completing the User-Centric Solution section, teams are strongly encouraged to first conduct two foundational exercises:

- Ideation

✅ **Tip:** Keep language user-focused, avoid technical jargon, and write in short, clear sentences.

Target Core User

Purpose: Ensure that your solution is user-centered and solves real user problems
When: By Week 3

Key tools:
- User Persona

Tip: Once you are Summarise the information

Targeted User Persona



Maria Reyes
Contract Specialist

Demographic info
Age 29, bachelor's degree in business administration, procurement certification, mid-sized city government

Short Bio

Maria has 8 years in public procurement. She handles 40–60 purchases a month, from office equipment to maintenance services. She is careful and rule-following, and she is often stretched thin at fiscal year-end.

Goals & Motivations

- *Award contracts on time*
- *Get the best value for taxpayers*
- *Pass audits without findings*

Frustrations & Pain Points

- *Hours spent researching prices*
- *Fear of missing a fraudulent or banned vendor*
- *Rewriting similar memos over and over*

Ideal Experience / Must Have

- *Upload quotes and get a clear risk summary in minutes*
- *Every flag shows its evidence*
- *She makes the final call*
- *A memo that's ready to file*

Skills & Capabilities

Expert in procurement rules; comfortable with Excel, email and government portals; not technical with AI

Solution Concept (Part 2)

Purpose: Understand what your solution will do and how it will solve the user pain-points
When: By Week 3

Key tools:
- Big Idea Vignettes
- To-Be-Scenario

Tip: …

Idea Cluster Play Back

1. ***Price intelligence:*** *compare quotes with past government prices and market prices*
2. ***Vendor integrity:*** *registration, exclusions list, links between vendors*
3. ***Bid pattern detection:*** *suspiciously close bids, contracts split to stay under approval limits*
4. ***Reporting and transparency:*** *officer memo and public summary*
5. ***Human control:*** ***a****ccept or override flags, award, approve publication*

Value Proposition & AI Justification

For procurement officers who must prove every purchase is fairly priced and fraud-free but research prices and vendors by hand, ClearBid is an agentic AI assistant that reviews quotes in minutes, flags risky or colluding vendors with evidence, and drafts audit-ready reports. AI is needed to read messy quotes, match products described differently, weigh evidence from several sources and write for two audiences. Fixed rule checks run as plain code tools.

Ealy Concept (Low Fidelity Prototype)

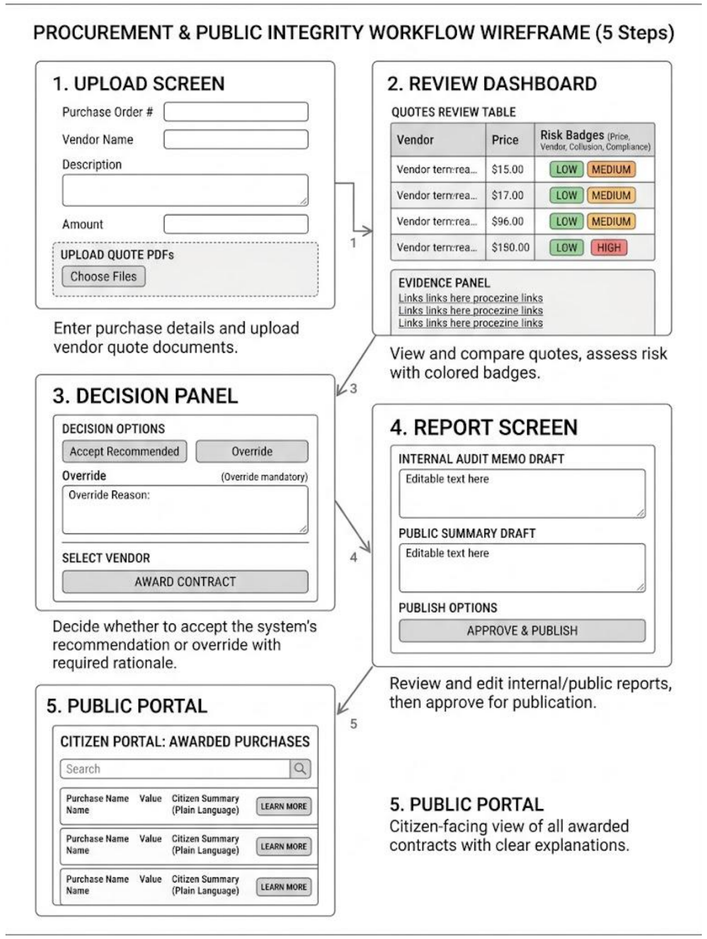
Upload screen: the officer enters the purchase details and uploads the quote PDFs

Review dashboard: a table of quotes with risk badges (price, vendor, collusion, compliance) and an evidence panel

Decision panel: Accept / Override (reason required) → Select vendor → Award

Report screen: draft memo and public summary → Approve publication

Public portal: citizen-friendly list of purchases with plain-language explanations



Innovation Intent

Key tools:
- Big Idea Vignettes
- To-Be-Scenario

Tip: …

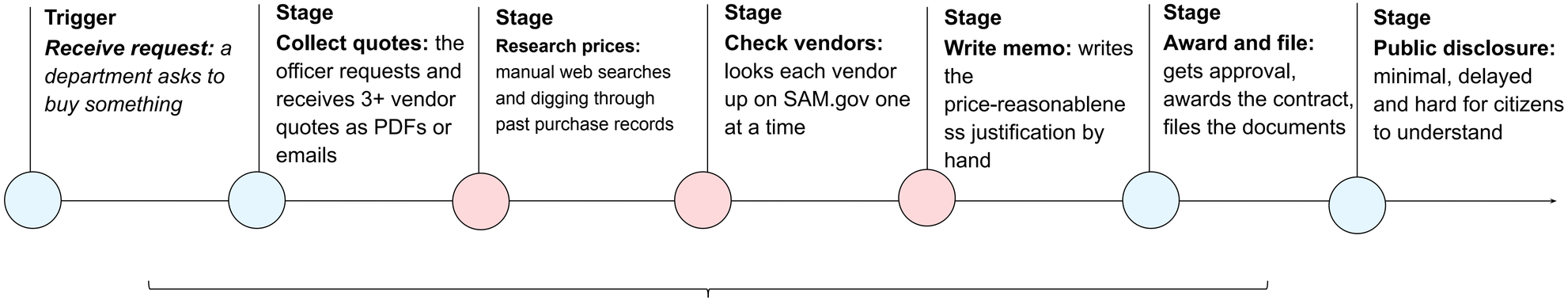
Purpose: Explain why this solution is different
When: By Week 3

User goal

Make fast, defensible and transparent purchase decisions.

Proposed transformation

from hours of manual research across separate websites to a single AI-assisted review in minutes, with evidence, human decisions and automatic public reporting.



Why this solution

it uses data the government already publishes, fits into the officer's existing workflow, and keeps humans accountable for every decision.

What makes this innovative?

- *New workflow: review, decide and publish in one connected flow instead of separate steps*
- *New logic: combines weak signals across vendors to detect collusion, not only single-vendor checks*
- *New coordination: a supervisor agent routes work between review and reporting agents*
- *New data use: turns raw USAspending and SAM.gov records into live price benchmarks*
- *Controlled autonomy: agents investigate on their own but cannot award contracts or publish*

Source: If applicable, describe source origin

✅ Checkpoint: Week 3

At this stage, you should have a clear picture of what your solution will look like – this is your first solution design.

- ❑ Is the concept clear?
- ❑ Is this solution different/innovative compared to what is already in the market?

Key tools:
- Ideation
- Concept prototyping

Tip: Early testing & validation …. Start getting use and expert feedback by showing your solution concept.

Open Questions & Assumptions

What we don't know yet

List gaps in knowledge that need validation.

1. *Whether officers prefer a risk score, flags, or both*
2. *How detailed the public summaries should be without exposing sensitive vendor data*

Assumptions we are making

Document any assumptions about users, processes, or technology.

1. *One combined review agent is enough for an MVP*
2. *Synthetic quotes and vendors can realistically represent real cases*

Validation research

List what research is needed to gather information and/or validate assumptions

- ❑ *1. Test the prototype screens with 2–3 classmates or a coach acting as officers*
- ❑ *2. Compare the draft memo against public price-reasonableness memo examples*

Solution Blueprint

From idea to reality –
building strong AI solutions

AI Agent Architecture

Purpose: Show the *structural design* of your agent – what major components it consists of and how they connect.

When: By week 4-5

Key tools:
- IBM Orchestrate

Tip: Architecture determines what your agent can do and how it does it.

✔ Use clear diagrams (e.g., flowcharts for workflow).
✔ Keep text concise and diagrams labelled.
✔ Be ready to explain each component and transition.

Agent Overview

ClearBid uses 3 agents built in watsonx Orchestrate: 1 supervisor and 2 specialists. Rule-based checks run as Python tools to keep results exact and model costs low.

Agent roles (single or multiple):

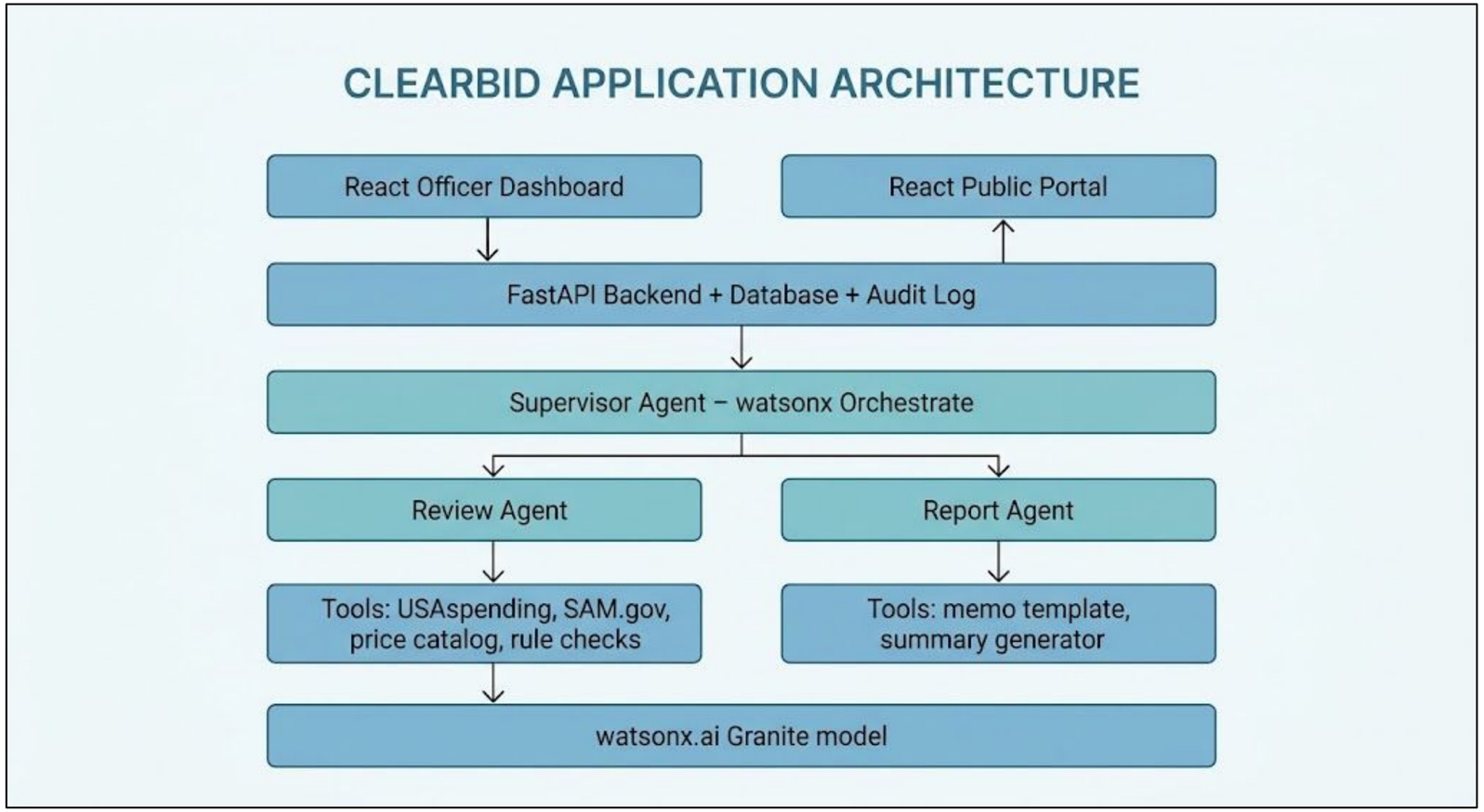
If your solution includes multiple agents, explain how they work together to generate the final output

Agent	Responsibility
Supervisor agent	receives the purchase request, sends work to the specialists, collects the results and returns them to the officer dashboard
Review agent	extracts quote details; compares prices with USAspending history and the market catalog; screens vendors (registration, exclusions, shared details); checks bid patterns and key procurement rules; and combines the evidence into flags
Report agent	drafts the internal price-reasonableness memo and the plain-language public summary

Why this architecture?

Low cost, Reliable, Safe and Extensible.

Architecture Diagram



AI Agent Workflow

Purpose: Show the *dynamic process* your agent follows to achieve its goals — step by step.

When: By week 4-5

Key tools:
- IBM Orchestrate

Tip: Ensure that the sequence clear and logical and that iterations and decision points are indicated indicated.

✔ Use clear diagrams (e.g., flowcharts for workflow).
✔ Keep text concise and diagrams labelled.
✔ Be ready to explain each component and transition.

Workflow Description

The officer uploads quotes. The supervisor sends them to the review agent, which runs the checks and returns flags with evidence. The officer decides. The report agent then drafts the memo and public summary, and the officer approves publication. Every step is logged.

Decision Points

- Is the vendor on the exclusions list? → mark as blocked, pending the officer's confirmation
- Is the price more than a set threshold above the benchmark (e.g., 25%)? → price flag
- Do competing vendors share an address, phone number or bank account? → collusion flag
- Is the purchase amount above the approval limit? → supervisor sign-off needed
- Are enough quotes provided for this amount? → compliance flag

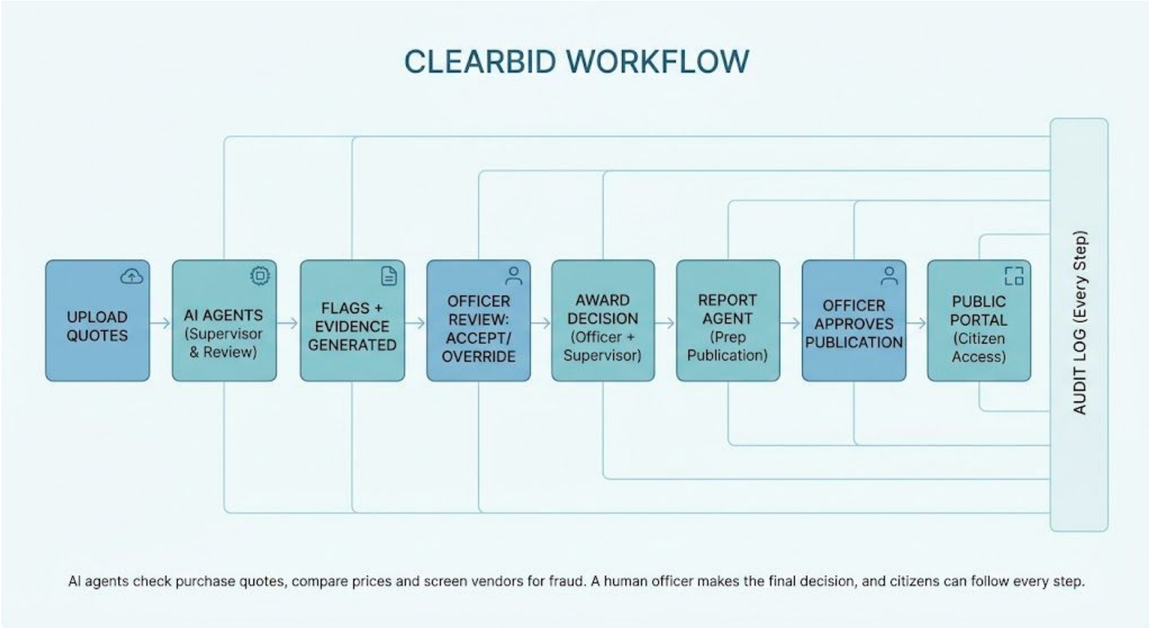
Human-in-the-loop moments

- The officer starts the request and uploads the quotes
- The officer accepts or overrides each flag (with a reason for any override)
- The officer awards the contract; a supervisor co-signs above the limit
- The officer approves the public summary before it's published
- An auditor reviews the decision log

Failure strategy nodes

- API down or slow: use cached data and label results "based on cached data from [date]"
- PDF can't be read: ask the officer to confirm or enter the line items manually
- No price benchmark found: mark "insufficient data" instead of guessing
- Model output invalid or unclear: retry once, then return to the officer for manual review
- Uncertain vendor name match: show it as "possible match – verify," never as confirmed

Workflow Diagram



Tools & Integrations

Purpose: List the external systems your agent uses — APIs, data sources, models, or libraries.

When: By week 4-5

Key tools:
- IBM Orchestrate

Tip: You can combine any supported external tool & integration to create and/or enhance your agentic AI solution. Keep in mind: Are the tools necessary and justified? How does integration affect performance?

Tools List & Purpose

Tool/Service	**Used for**	**Integration Type**	**Executables vs autonomy** *Invoked by agents or workflows* *Automated action or user-verified step*
watsonx Orchestrate	Builds and runs the 3 agents	IBM platform	Runs agents; routes autonomously
watsonx.ai	Reasoning, product matching, report writing	Model API via Orchestrate	Recommends only
IBM Bob	Writes the React app, FastAPI tools, specs, test data	AI coding assistant (development)	Developer reviews all code
USAspending.gov API	Past government purchase prices	REST API (free, no key), cached	Read-only
SAM.gov API	Vendor registration and exclusions	REST API (free key), cached	Read-only
Mock price catalog	Market price comparison	Local JSON / database	Read-only
FastAPI tools	Price math, vendor checks, bid patterns, rules	OpenAPI tools imported into Orchestrate	Runs automatically, results are exact
React app	Officer dashboard and public portal	Web frontend	Humans decide here
SQLite / PostgreSQL	Requests, flags, decisions, audit log	Database	Log can’t be edited

Notes on Constraints

- *Limited IBM credits: 3 agents, short prompts, a small Granite model, and Python for every exact check*
- *API responses are cached to avoid rate limits and repeated calls*
- *Quotes, vendors and the retail catalog are synthetic and clearly labelled*
- *No live web scraping (unreliable, and it may break website terms of service)*

Optional

Intelligence & Reasoning Design

Purpose: Define the technical depth – the brains of the agent apart from just architecture and workflow.

When: By week 5

Knowledge & Retrieval

- *USAspending award history*
- *SAM.gov vendor and exclusion records*
- *The mock price catalog, and a short rules file (quote requirements and approval limits)*

Reasoning & Decisions

- Python tools produce exact signals.
- The review agent weighs the signals together, sets a risk level (low, medium or high) and explains why in plain language.

Multi-Agent Logic (if used)

- The supervisor routes the work.
- The review agent finishes before the report agent starts.
- The report agent only uses decisions the officer has approved.

Risks & Constraints

- *Wrong product matches*
- *False fraud flags on honest vendors*
- *Vendor name confusion*
- *Limited model credits.*
- *These are handled by showing the evidence, labelling confidence and requiring a human decision.*

Governance & Ethics

Purpose: Explain how your AI agent is designed to behave *safely, responsibly, ethically* and in compliance with privacy and governance expectations. Include mechanisms that *prevent harm*, *protect users*, and *build trust*.

When: by Week 5

Ethical Considerations

Show how the agent is designed to respect rights, fairness, and human values

a) Fairness & Anti-bias

- *every vendor gets the same checks. Flags are based on evidence, not on company size or location. Flags don't automatically disqualify anyone.*

b) Transparency & Explainability

- *every flag shows its data source and reasoning, and citizens see plain-language summaries.*

d) Respect for Rights

- *vendors aren't rejected by AI. Officers record a reason for every decision, which supports appeals and bid protests.*

e) Privacy& Security

- *personal and sensitive vendor data (bank details, contacts) is removed from public reports.*

c) Accountability & Human Oversight

- *a named officer approves every decision, and the audit log can't be edited.*

Governance & Compliance

Explain how the design aligns with internal and external policies and frameworks:

- *Agents have no tools to award contracts or publish*
- *The workflow is aligned with FAR price analysis guidance (Part 15.404-1) and the SAM.gov exclusions requirement*
- *Each decision step is enforced in backend code*
- *Model prompts and versions are logged for audits*
- *watsonx.governance is possible future work for model monitoring*

Scope & Limitations of the MVP

Purpose: Explain what your MVP will and will not do.
When: by Week 5

Tip: At the end, you only have 10 weeks. You will need to prioritise – focus on the 2-3 functionalities that will deliver the greatest user value for your MVP.

MVP Scope

- Upload and review quotes for a single purchase (the 50-laptop demo)
- Price comparison against USAspending data and a mock catalog
- Vendor checks: exclusions list, shared address, phone or bank account
- 2 bid pattern checks and 3 procurement rules
- Officer dashboard with accept/override, award and approve
- Draft memo and public summary
- Public portal and audit log

Limitations

- *Synthetic quotes, vendors and market prices*
- *No live web price search*
- *Federal rules simplified; no state or local rule sets*
- *No login or role management beyond the basics*
- *English only*

What is next

- *Real market price APIs*
- *More fraud patterns (contracts split across several purchases, rotating winners)*
- *Integration with agency procurement systems*
- *State and local rule packs*
- *watsonx.governance monitoring*
- *Multi-language public portal*

✅ Checkpoint – Week 5

Tip: Validate technical decisions early with prototypes and stress tests. Ensure ethical and governance considerations are not an afterthought—they should be integrated into design from the start.

At this stage, you should be clear about the MVP that you are building and will submit at the end of the this program.

- ❑ Is the architecture modular, scalable, and aligned with solution goals? Does it supports performance, scalability, and maintainability?
- ❑ Are workflows clear, efficient, and user-centric? Does it aligns with user experience and business objectives?
- ❑ Are tools and integrations secure, compitble (interoperable), and cost-effective?
- ❑ Is the intelligence logic well-though and justified?
- ❑ Does the design ensures data privacy, fairness, explainability, and bias mitigation?
- ❑ Does the governance framework addresses compliance, and accountability?

Open Questions & Assumptions

What we don't know yet

List gaps in knowledge that need validation.

1. *How accurate product matching is against real USAspending descriptions*
2. *How many model credits one full review uses*

Assumptions we are making

Document any assumptions about users, processes, or technology.

1.
2. *3 agents and a small Granite model are enough for demo-quality results*
3. *Orchestrate can call the FastAPI tools reliably through OpenAPI*

Validation research

List what tests are needed to gather information and/or validate assumptions

- ❑ *1. Measure credit usage per review in a test run*
- ❑ *2. Run the demo scenario plus 2 variations (one clean, one fraudulent) to check accuracy*
- ❑ *3. Have a coach or peer review the memo for clarity*
- ❑ *…*

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Continue the journey

- ***Weeks 6-8:***
  - *Implement the technical work*
  - *Ensure depth & robustness*
  - *Carry out user testing & iteration*

- ***Weeks 9-10:*** *Prepare and finalise your project submission*

Good Luck!!